

Theorem (Markov's inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ be a non-negative random variable. For any $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Equality holds iff X takes only the two values 0 and a almost surely (equivalently $X = a \mathbf{1}_{\{X \geq a\}}$ a.s.).

Proof. 1. Assumptions and degenerate case. If $\mathbb{P}(X \geq a) = 0$, the left-hand side of the desired inequality is 0, while $\mathbb{E}[X] \geq 0$ and $a > 0$ imply $\mathbb{E}[X]/a \geq 0$. Hence the inequality holds trivially, and we may assume $\mathbb{P}(X \geq a) > 0$.

2. Indicator of the event $\{X \geq a\}$. Define

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

3. Pointwise bound. For every $\omega \in \Omega$ we have

$$a \mathbf{1}_{\{X \geq a\}}(\omega) \leq X(\omega). \quad (1)$$

Indeed, if $X(\omega) \geq a$ then the left-hand side equals $a \leq X(\omega)$; if $X(\omega) < a$ the left-hand side is $0 \leq X(\omega)$ by non-negativity of X .

4. Taking expectations. The expectation operator is monotone, so applying it to (1) gives

$$a \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}[X]. \quad (2)$$

5. Expectation of an indicator. By definition,

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (3)$$

Substituting (3) into (2) and dividing by the positive constant a yields

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is Markov's inequality.

6. Equality condition. Equality in the final inequality can occur only if every inequality used above is an equality almost surely.

- Equality in the monotonicity step (2) forces the pointwise bound (1) to hold with equality $\mathbb{P}$ -a.s., i.e.

$$a \mathbf{1}_{\{X \geq a\}}(\omega) = X(\omega) \quad \text{for } \mathbb{P}\text{-a.e. } \omega.$$

- This identity means $X(\omega) = a$ on the set $\{X \geq a\}$ and $X(\omega) = 0$ on its complement.

Thus X must be of the form $X = a \mathbf{1}_{\{X \geq a\}}$ a.s., i.e. X takes only the values 0 and a . Conversely, for such a two-point distribution we have $\mathbb{E}[X] = a \mathbb{P}(X = a)$, so $\mathbb{P}(X \geq a) = \mathbb{E}[X]/a$ and equality holds.

7. Conclusion. For every non-negative random variable X and every $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

with equality precisely when X assumes only the values 0 and a almost surely.

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